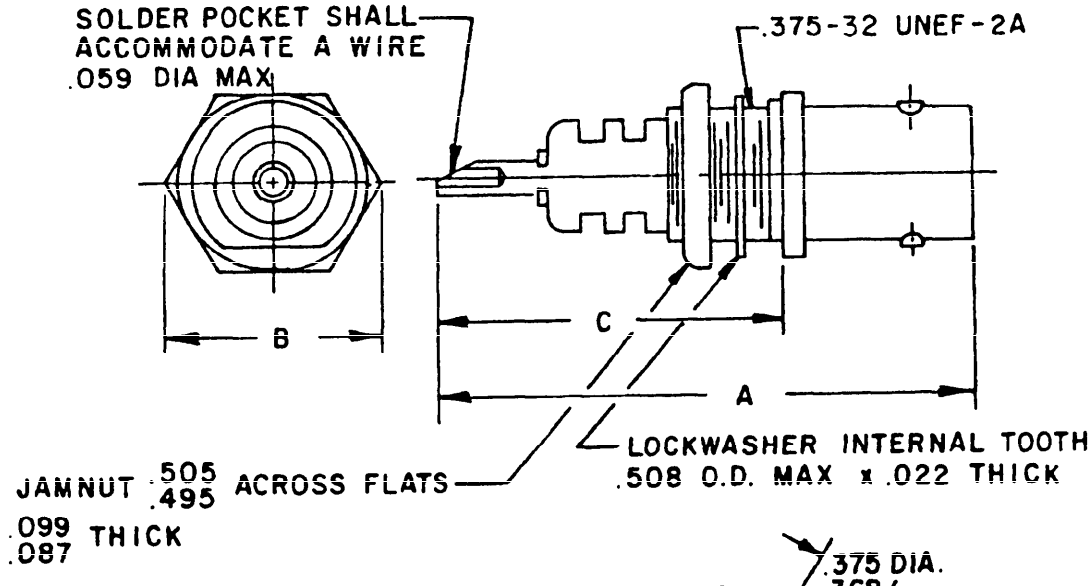
MILITARY SPECIFICATION SHEET

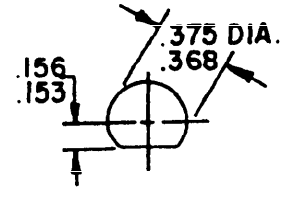
CONNECTORS, COAXIAL, RADIO FREQUENCY, HIGH VOLTAGE (SERIES MHV) (UNCABLED) - RECEPTACLES, SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

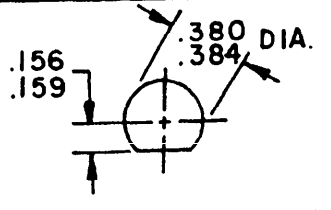
The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-39012.

INACTIVE FOR NEW DESIGN





CONNECTOR D MTG SURFACE



MOUNTING HOLE

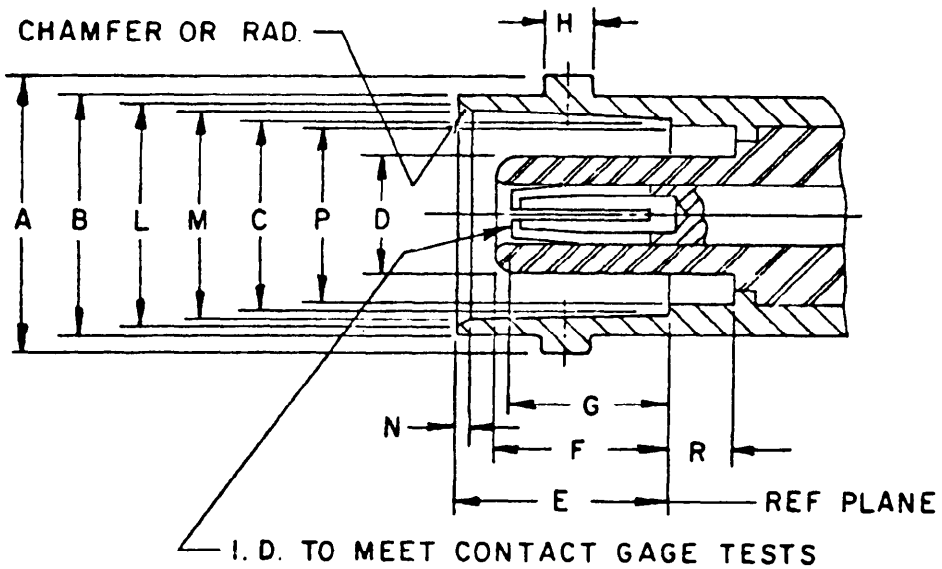
INCHES	MM	INCHES	MM
.022	.56	.368	9.35
.059	1.50	.375	9.52
.087	2.21	.380	9.65
.099	2.51	.384	9.75
.153	3.89	.429	10.90
.156	3.96	.434	11.02
.159	4.04	.439	11.15
.181	4.60	.443	11.25
.184	4.67	.495	12.57
.187	4.75	.505	12.83
.190	4.83	.508	12.90

NOTES:
1. Dimensions are in inches.
2. For dimensions 'A', 'B' and 'C' see table I.
3. Dimension 'B' is the largest overall diameter of the connector.
4. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
5. Orientation of mounting flats and bayonet studs shall be within ±3° of the orientation shown.
6 Allundimensioned pictorial representations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

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LTR	Dimensions in inches with metric equivalents (mm) in parentheses. Minimum	Maximum
A	.432 (10.97)	.436 (11.07)
B	.378 (9.60)	.382 (9.70)
C	.319 (8.10)	.321 (8.15)
D		.186 (4.72)
E	.327 (8.31)	.335 (8.51)
F	.289 (7.34)	.311 (7.90)
G	.253 (6.43)	.280 (7.11)
H	.075 (1.91)	.081 (2.06)
J	.270 (6.86)	
K	.081 (2.06)	.091 (2.31)
L	.346 (8.79)	.356 (9.04)
M	.327 (8.31)	.333 (8.46)
N	.015 (.38)	.030 (.76)
P	.284 (7.21)	.290 (7.37)
R	.086 (2.18)	

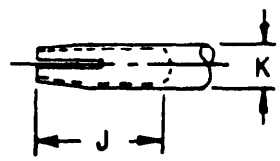


FIGURE 2. Mating dimensions for socket terminations.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: Not applicable.

Voltage rating:
1600 volts rms, maximum working voltage at sea level.
375 volts rms, maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - 3 pounds, maximum.
Torque - 2-1/2 inch-pounds, maximum.

Mating characteristics:
See figure 2 for dimensions.
Outer contact:
Test ring ID - .319 maximum, 16 microinch finish.
Insertion force - 5 pounds, maximum when inserted a minimum of .093.
Contacts with slotted members: Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 5,000 megohms, minimum.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact	2.0	2.5
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 5,000 volts rms, minimum at sea level.

Corona level:
Voltage - 1245 volts rms.
Altitude - 70,000 feet.

RF high potential withstanding voltage: Not applicable.

Connector durability:
500 cycles minimum at 12 cycles/min maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.
Initial - 5 pounds, maximum.
Final - 5 pounds maximum; 1 pound, minimum.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202. No discontinuities.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I.
Acceleration - 50 G's at 7 milliseconds. No discontinuities.

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TABLE I. Part number and overall dimensions.

Part number	Dim	Inches-millimeters		Panel thickness	
		Minimum	Maximum	Minimum	Maximum
M39012/104-					
0001	A		1.65 (41.91)		
	B		.585(14.86)	.045(1.14)	.260(6.60)
	C	.843 (21.41)	.990(25.15)		

Thermal shock: Method 107, test condition B, MIL-STD-202, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cable (see table I).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Cable retention force:
Noncrimp assemblies - 40 pounds, minimum.

Coupling mechanism retention force: Not applicable.

Part number: M39012/104-0001

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AM, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3065-5)

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